

Saumya Pratap Singh

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Education

United College of Engineering and Research | *B.Tech in CSE (CGPA: 8.2/10)* 2023 – 2027
• **Coursework:** Data Structures & Algorithms, AI, DBMS, OOPs with Java, Full Stack Web Development, OS, DAA.

Experience

Centre for Advanced Studies | *Rapid Prototyping and Design Thinking Intern* Aug 2025 – Sep 2025
• Spearheaded **CleanSweep AI**, a data-driven platform linking citizens, authorities, and recyclers.
• Designed system architecture to visualize real-time IoT data, optimizing urban waste collection routes.

IBM | *Software Engineering Intern* Sep 2025 – Nov 2025
• Developed scalable web applications using the **MERN stack**, ensuring robust backend performance.
• Architected secure RESTful APIs to handle user authentication and payment processing.

1M1B (1 Million for 1 Billion) | *Data Analyst Intern* June 2025 – July 2025
• Contributed to environmental sustainability projects by analyzing large datasets using **Tableau**.
• Generated actionable insights from raw data to drive strategic decision-making for the 'Green Initiative'.

Projects

Kumbh-Rakshak: AI Surveillance System | *FastAPI, DeepFace, React.js, PostgreSQL, Tailwind* [GitHub](#)
• Built an **Intelligent Biometric Grid** using **DeepFace** & **Supabase (pgvector)** to identify persons in **500ms**.
• Engineered a **Geospatial Tracking Dashboard** with Leaflet.js to visualize suspect trajectories from edge nodes.

OpenBank Pro | *MERN Stack (MongoDB, Express, React, Node), JWT, Redux* [GitHub](#)
• Architected a secure banking simulation with **Role-Based Access Control** and **Anti-Money Laundering** checks.
• Implemented a complex **UPI Payment Module** and utilized **node-cron** to automate recurring deposits.

CleanSweep AI | *IoT (Arduino), Gemini API, Supabase, Leaflet.js, Real-time DB* [GitHub](#)
• Developed a smart waste management system integrating **IoT Sensors** with a real-time **Heatmap Visualization**.
• Leveraged **Google Gemini API** to process uploaded images, validating cleanliness reports with **95% accuracy**.

Krishi Seva: AI Farmer Companion | *Flutter, Dart, TensorFlow Lite, Firebase, Hive* [GitHub](#)
• Developed an offline app using **TensorFlow Lite (EfficientNetB0)** to detect plant diseases with **96.3% accuracy**.
• Bridged the digital divide by integrating **Government Schemes** and **Hindi localization** for rural accessibility.

Technical Skills

Languages: C++, Python, JavaScript, SQL, Dart, Java, HTML/CSS
Frameworks: React.js, Node.js, Express.js, FastAPI, Flutter, Tailwind CSS, Bootstrap, Redux
AI/ML: TensorFlow Lite, DeepFace, OpenCV, Google Gemini API, LangChain, NumPy, Pandas
Cloud & Tools: Microsoft Azure, Oracle Cloud, Google Cloud Platform, Docker, Postman, Git, GitHub, Supabase
Core Concepts: REST APIs, Vector Databases, System Design, IoT Architecture, Microservices, Data Structures

Certifications

1. Oracle Cloud Infrastructure Certified Associate | 2. Postman API Student Expert | 3. Google Prompting Essentials | 4. Google Analytics | 5. IBM Web & Mobile Dev | 6. Cyber Security Foundation | 7. AI for Beginners: HP

Achievements & Leadership

- **Hackathon Winner:** TechSprint National Level Hackathon at Ujjain Engineering College.
- **Winner:** Kaggle Community Olympiad - Unmasking Fakes (Organized by TFUG Prayagraj).
- **Secured Top 3 Rank:** GeeksforGeeks DSA College Contest (Week 2 & Week 3).
- **Hackathon Finalist:** Zinnovatio 2.0 Hackathon (Chandigarh University).
- **Ideathon Qualifier:** UHack 3.0 Ideathon (United College of Engineering and Research).
- **Leadership:** Core Member (GDG & TFUG Prayagraj) | Ambassador (Google Gemini, Unstop, Girl-Script).
- **Volunteer:** NSS (National Service Scheme) | Turbo AI Volunteer (LMT Fest AKTU 2024-25).